Synoptic LE: Tree Fluxes 2023

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2026-04-23

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1 Read in and collate all the files in the Processed Data folder

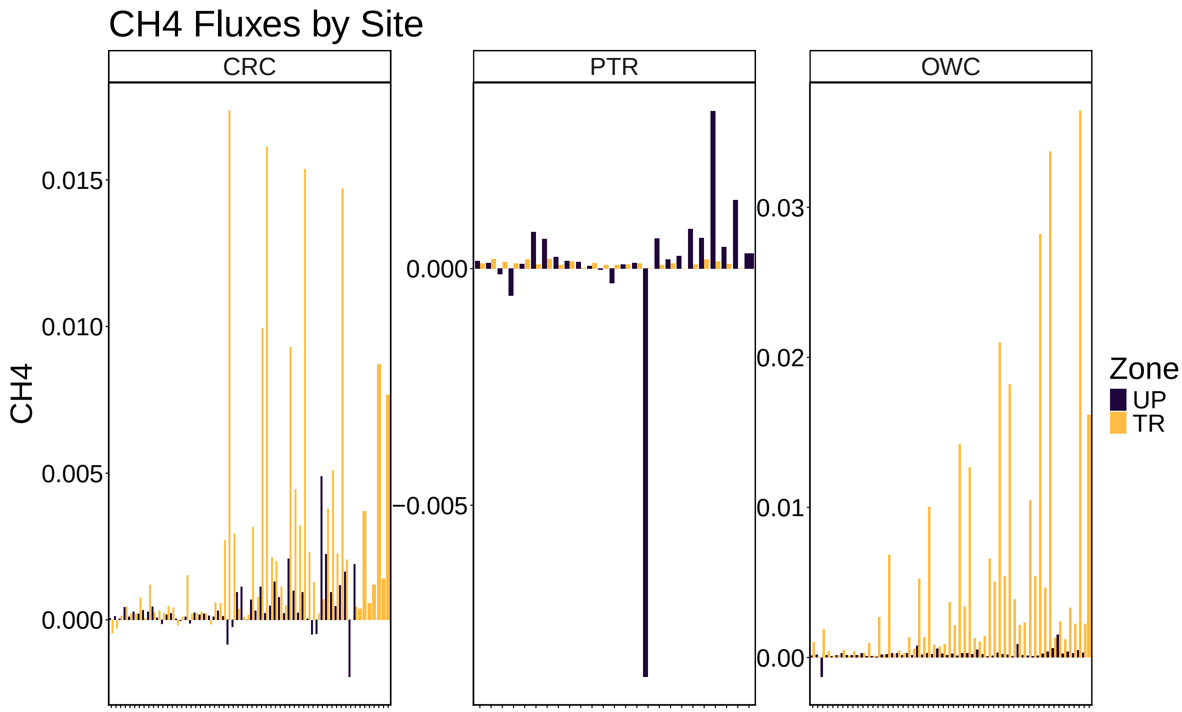
2 Remove bad fluxes and calculate how many there were

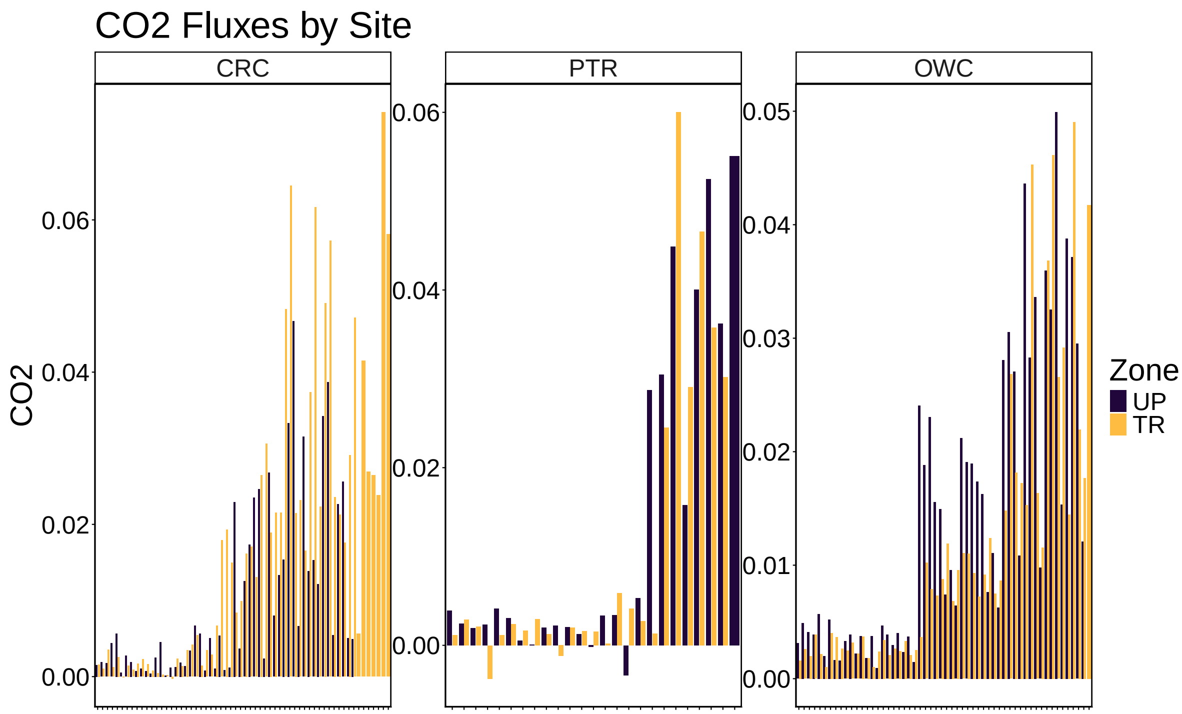
```
##Remove any samples with "remove" in the qaqc_flag columns
final_dat <- all_dat %>%
  filter(!grepl("remove", ch4_qaqc_flag, ignore.case = TRUE)) %>%
  filter(!grepl("remove", co2_qaqc_flag, ignore.case = TRUE))

##Removed fluxes in the CH4 and CO2 flux processing Rmd code, removed both CH4 and CO2 obs wh
```

```
##note that this list has some not synoptic monitoring samples zone-labeled as "Luci" that we

##note that this list has some diurnal variation samples for PTR synoptic veg campaign in 202
```

3 Write out files

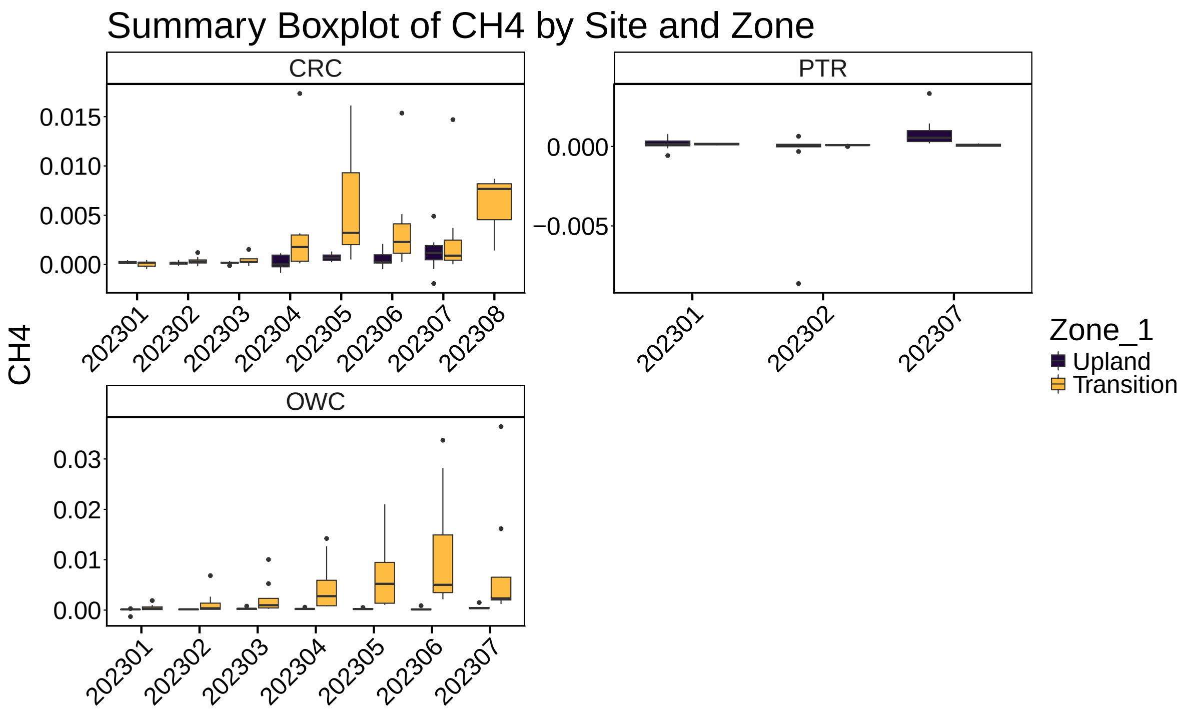
4 All Data by Year, Site and Zone

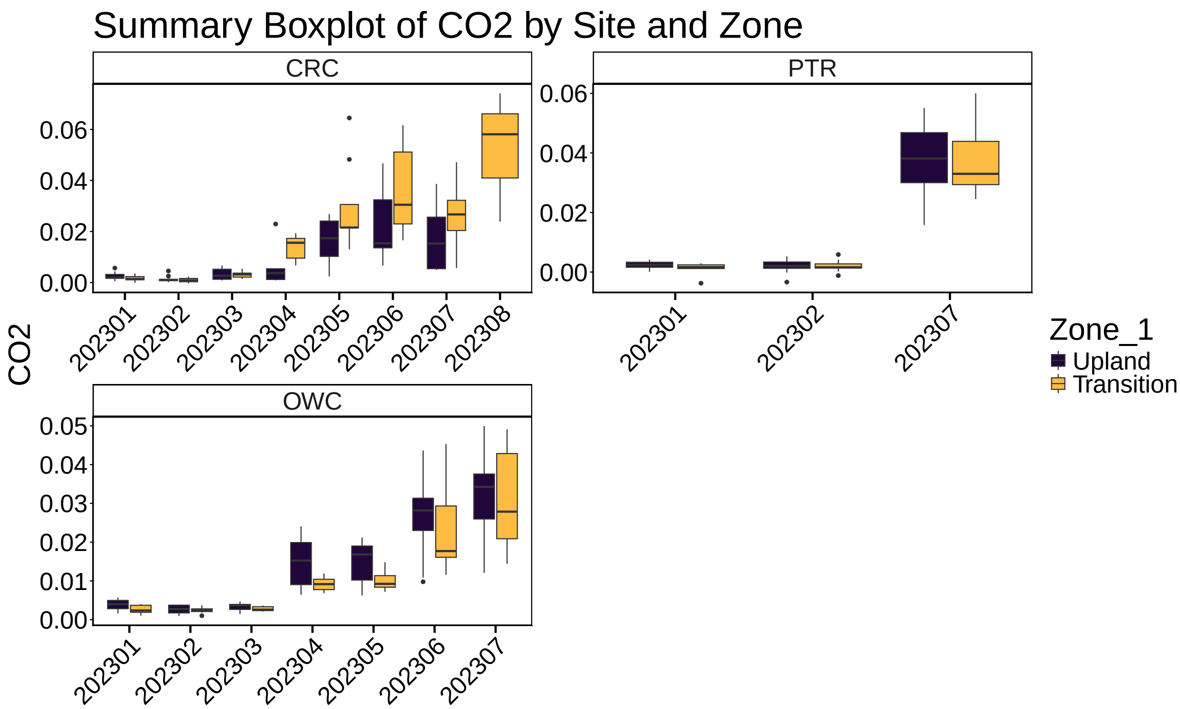




CO2 Fluxes by Site
CRC
PTR
OWC
CO2
0.06
0.04
0.02
0.00
0.06
0.04
0.02
0.00
0.05
0.04
0.03
0.02
0.01
0.00
Zone
UP
TR

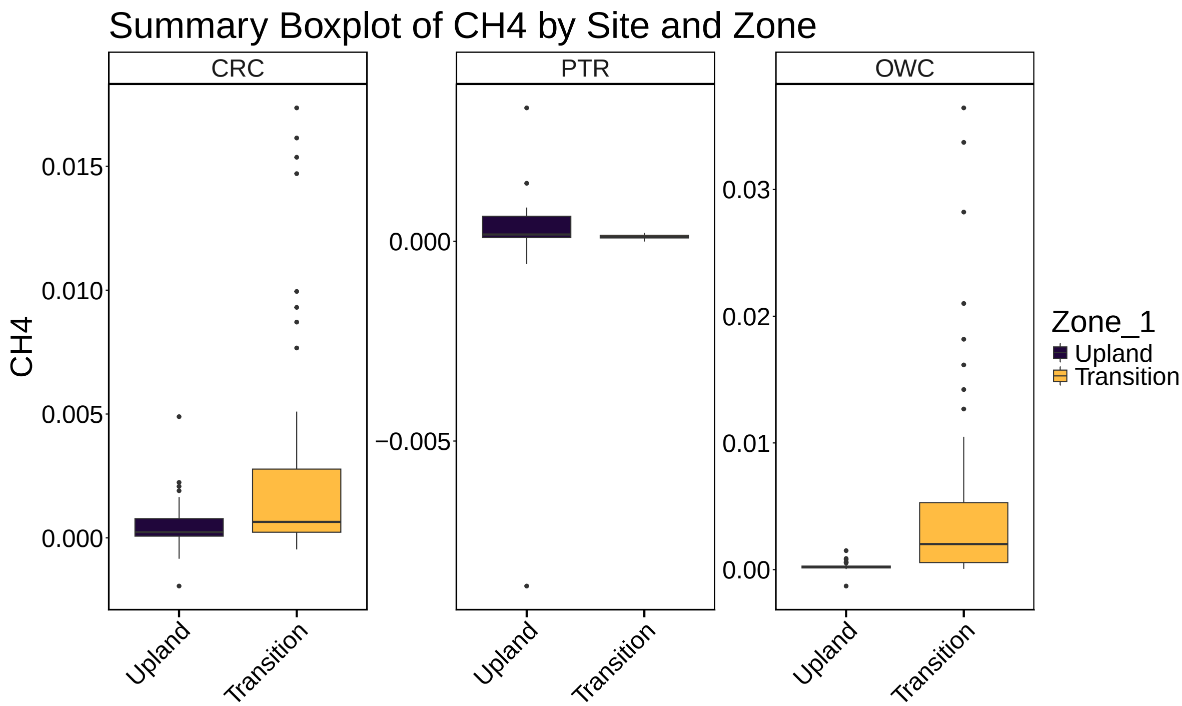
5 Boxplot summary by Date

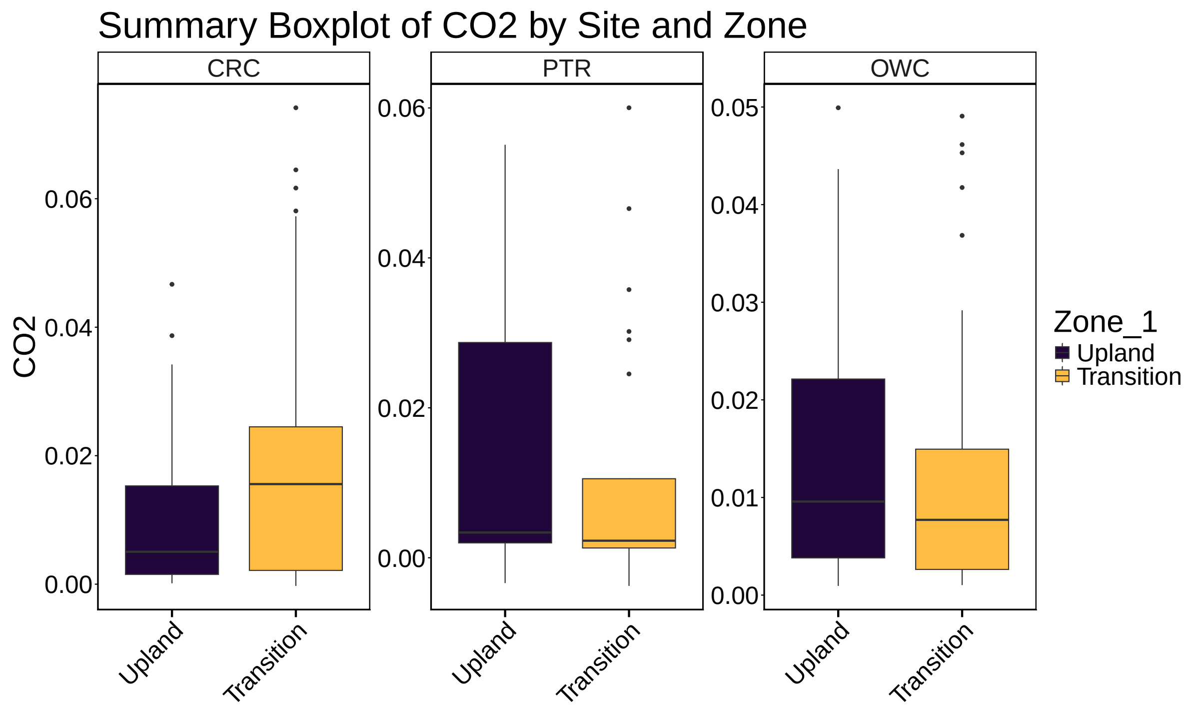




Summary Boxplot of CO2 by Site and Zone
CRC
PTR
OWC
CO2
0.06
0.04
0.02
0.00
0.05
0.03
0.01
202301
202302
202303
202304
202305
202306
202307
202308
Zone_1
Upland
Transition

6 Boxplot summary by Site





7 Calculate summary metrics

8 Summary Tables

Table 1: Summary Statistics

Site	Zone_1	min_ch4	max_ch4	mean_ch4	min_co2	max_co2	mean_co2
CRC	Upland	-0.0019429	0.0048932	0.0004683	0.0001376	0.0466813	0.0100528
CRC	Transition	-0.0004658	0.0173559	0.0025957	-0.0002721	0.0741587	0.0180612
PTR	Upland	-0.0086278	0.0033354	0.0000446	-0.0033748	0.0550926	0.0135276
PTR	Transition	-0.0000081	0.0002088	0.0001052	-0.0037544	0.0600224	0.0106762
OWC	Upland	-0.0012904	0.0015022	0.0002408	0.0009411	0.0499195	0.0141066
OWC	Transition	0.0000663	0.0364326	0.0051798	0.0010240	0.0490653	0.0116198